

{EPITECH}

BOOTSTRAP

< MY_LS />



BOOTSTRAP



binary name: info_file

language: C

compilation: via Makefile, including re, clean and fclean rules

You must develop a program that takes a file name as parameter and displays the following information:

- ✓ File name
- ✓ File type
- ✓ Inode
- ✓ Number of hard links
- ✓ File size
- ✓ Allocated space on hard drive
- ✓ Minor (if file is "device")
- ✓ Major (if file is "device")
- ✓ Owner's UID
- ✓ Owner's GID

Authorized Functions

The only system calls allowed are the following ones:

- ✓ opendir, readdir, closedir, stat, lstat, write
- ✓ getpwuid, getgrgid, time, ctime, readlink,
- ✓ malloc, free, perror, strerror, exit.



Read the manual first

man 1 stat

man 2 stat

man 3 stat

Examples

```
Terminal
$> ./info_file /bin/ls
Name: ls
Type: f
Inode: 424242
Hard Link: 2
Size: 110080
Allocated space: 112000
Minor: N/A
Major: N/A
UID: 0
GID: 0
```

```
Terminal
$> ./info_file /dev/tty
Name: tty
Type: c
Inode: 17
Hard Link: 1
Size: 0
Allocated space: 0
Minor: 0
Major: 5
UID: 0
GID: 5
```

```
Terminal
$> ./info_file /dev/toto
File not found
$> echo $?
84
```

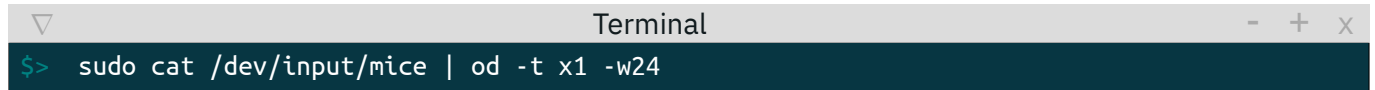
```
Terminal
$> ./info_file path/to/a_symbolic_link
Name: a_symbolic_link
Type: l
Inode: 22546188
Hard Link: 1
Size: 7
Allocated space: 4096
Minor: N/A
Major: N/A
UID: 1000
GID: 1000
```

Everything is a file

In Unix, device files are special files as they act as an interface between the operating system (and its user processes) and the hardware devices.

They basically let you treat hardware like files, so reading/writing to `/dev/sda` or `/dev/tty` is really talking to a disk or a terminal, not just a text file.

Let's see it in action and display the events coming from your mouse/trackpad.

A terminal window titled "Terminal" with standard window controls (minimize, maximize, close) on the right. The command prompt shows a green prompt character followed by the command: `sudo cat /dev/input/mice | od -t x1 -w24`.

```
Terminal
$> sudo cat /dev/input/mice | od -t x1 -w24
```

Now move your cursor.

We can see this file isn't made to be read by humans but by programs that need to handle mouse input (X server, Wayland compositors).

v 2.3

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